

CARISURG PROJECT BASELINE MODEL REPORT

Explainable Machine Learning for Emergency Department Triage

I. Objective and Dataset

The objective of this study was to develop and evaluate simple, explainable machine learning models capable of predicting Emergency Severity Index (ESI) levels using information available during Emergency Department (ED) triage. The aim was not to develop the highest-performing model, but rather to establish a transparent and clinically defensible baseline against which future models can be compared.

The Yale ED dataset used in this project contains 55,121 adult patient encounters classified into the five ESI triage levels. As shown in *Figure 1.*, the dataset is highly imbalanced, with ESI Levels 2 and 3 accounting for most presentations while ESI Level 1 represents only a very small proportion of patients. This imbalance creates a challenging classification problem because a model may achieve high overall accuracy simply by correctly predicting the most common classes while still missing critically ill patients.

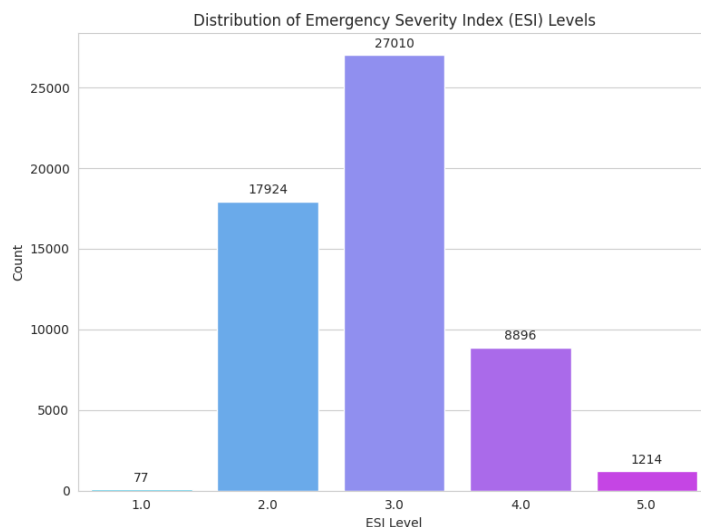


Figure 1. *Dataset Class Distribution.*

Only information available during the initial triage assessment was included to prevent data leakage. A total of **207 predictor variables** were used, comprising **200 binary chief complaint indicators** and **7 routinely collected triage vital signs** (heart rate, systolic/diastolic blood pressure, respiratory rate, oxygen saturation, temperature and blood glucose). Variables generated after triage, such as patient disposition, were deliberately excluded because they would not be available when the triage decision is made. The dataset was divided using a stratified **80:20 train/test split** with a fixed **random seed (42)**, ensuring that each ESI level maintained the same distribution in both datasets.

II. Baseline Model Development

Three baseline classifiers were evaluated. A **Dummy Classifier** provided a stratified random benchmark to demonstrate the minimum level of performance expected without learning meaningful clinical patterns. **Logistic Regression** was selected as an interpretable statistical baseline because its coefficients allow clinicians to examine which clinical features contribute most strongly to each prediction. A **Decision Tree** was also trained to provide a transparent flowchart of sequential clinical decisions. To reduce overfitting while maintaining interpretability, the maximum tree depth was limited to three levels.

III. Model Performance

All machine learning models outperformed the Dummy Classifier, demonstrating that they learned useful patterns from the triage data. Logistic Regression had the strongest overall performance across the evaluation metrics, shown in *Table 1.*, while the Decision Tree produced lower predictive performance but remained highly interpretable.

Model	Accuracy	Macro F1	Weighted F1	Recall (ESI 1)
Dummy Classifier	0.375	0.204	0.375	0.00
Logistic Regression	0.662	0.461	0.656	0.12
Decision Tree	0.518	0.168	0.383	0.00

Table 1. *Baseline Model Performance Comparison.*

The performance comparison shown in *Figure 2* highlights that **Logistic Regression** produced the strongest overall baseline performance, while the Decision Tree offered greater interpretability at the expense of predictive accuracy. Macro F1 treats all five ESI levels equally, whereas Weighted F1 gives greater influence to the more common ESI levels; the lower Macro F1 values therefore indicate weaker performance on the rare ESI Level 1 and Level 5 patients despite stronger overall performance on the majority classes.

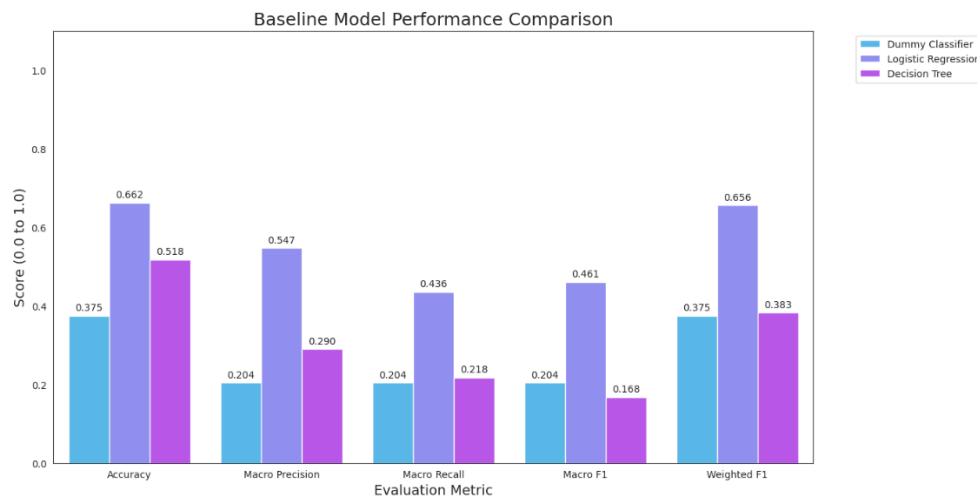


Figure 2. *Baseline Model Performance Comparison.*

IV. Clinical Interpretation and Metric Justification

Overall accuracy was *not* selected as the primary evaluation metric because the dataset is highly imbalanced, with most patients belonging to ESI Levels 2 and 3. A model can achieve good overall accuracy while still failing to identify the rare but critically ill ESI Level 1 patients. For this reason, **Recall for ESI Level 1** was considered the most clinically meaningful measure of model performance in this study.

Recall measures how many patients requiring immediate care are correctly identified by the model. In emergency triage, missing a critically ill patient is a greater clinical risk than incorrectly assigning a higher priority to a stable patient, as delayed treatment may have serious consequences. Prioritising Recall for ESI Level 1 therefore better reflects the clinical goal of identifying patients who need immediate intervention.

The confusion matrices (Figure 3) show that the baseline models performed reasonably well for the common ESI categories but struggled to identify the rare ESI Levels 1 and 5.

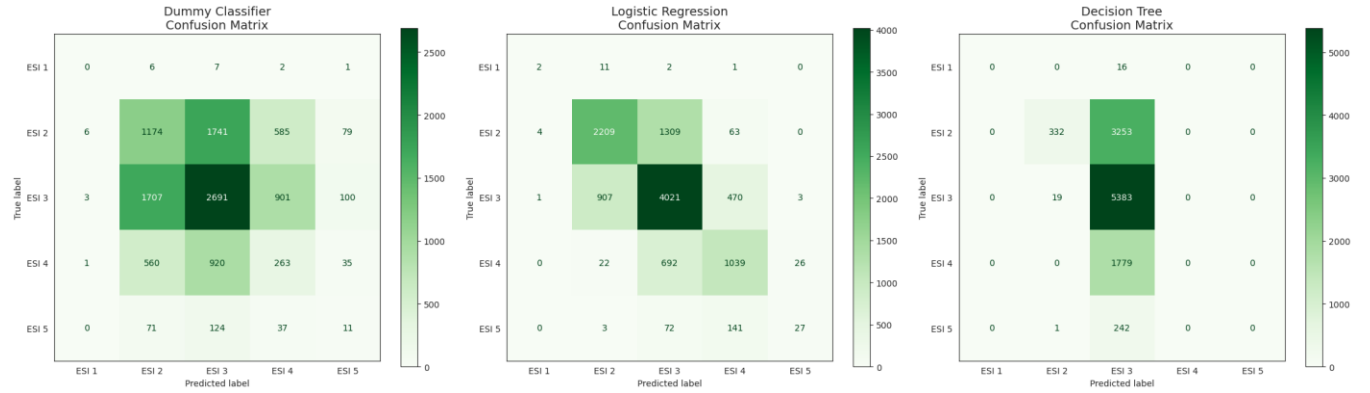


Figure 3. Confusion Matrices of the Three Baseline Models.

V. Failure Modes and Future Work

Failure Modes

- Poor identification of ESI Level 1 patients due to the small number of examples available for training.
- Some critically ill patients may present with less obvious symptoms or physiological measurements, making accurate classification more difficult.
- Missing an ESI Level 1 patient could delay urgent assessment and treatment, increasing the risk of adverse patient outcomes.

Future Work

- Improve performance for minority classes using class balancing and hyperparameter optimisation.
- Evaluate more advanced machine learning models while maintaining model transparency for clinical decision support.

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